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TA 3 - DOCUMENTATION

Case Based Reasoning Mathematics Intelligent Tutoring System

The case base contains four previous student cases involving different mathematics topics and repeated learning errors. Each case stores the mathematics topic, the student's error pattern, submitted solution, successful intervention, and learning outcome. The cases include quadratic equations, linear equations, and factoring. These previous cases serve as the system's knowledge and experience when assisting a new student.

Similarity Method

The system compares the new student's problem with every case in the case base. It checks two attributes: the mathematics topic and the repeated error pattern. One point is given when the topic matches, and another point is given when the error pattern matches. The maximum similarity score is 2. The case with the highest score is retrieved as the most relevant previous student experience.

Adaptation Logic

After retrieving the most similar case, the system creates personalized feedback based on the new student's repeated error. For example, if the student is misusing the quadratic formula, the system recommends identifying the values of a, b, and c before substituting them into the formula. If the student has an arithmetic error, the system recommends solving one operation at a time and checking each step. In this way, the previous intervention is adapted to match the new student's specific learning struggle.

Retention and Learning

After the personalized feedback is provided, the user selects the student's outcome: "Improved" or "Needs more practice." The system then adds the new student's topic, error, submitted solution, feedback, and outcome to the case base. This demonstrates the retention stage of Case-Based Reasoning because the system stores the new experience for possible use in future student problems.

SCREENSHOTS

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MATHEMATICS INTELLIGENT TUTORING SYSTEM
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Choose a math topic:
1. Quadratic Equations
2. Linear Equations
3. Factoring
Enter your choice: 1

Choose the repeated error:
1. Misusing the quadratic formula
2. Wrong substitution
3. Arithmetic error
4. Incorrect factors
Enter your choice: 1

Enter the student's submitted math solution: I used b instead of -b in the formula.

Student outcome:
1. Improved
2. Needs more practice
Enter your choice: 1

CASE-BASED REASONING RESULTS
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NEW STUDENT PROBLEM
Math Topic: Quadratic Equations
Repeated Error: Misusing the quadratic formula
Submitted Solution: I used b instead of -b in the formula.

RETRIEVED SIMILAR CASE
Previous Topic: Quadratic Equations
Previous Error: Misusing the quadratic formula
Previous Solution: Used b instead of -b in the formula.
Previous Intervention: Review the quadratic formula and identify a, b, and c.
Similarity Score: 2/2
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ADAPTED PERSONALIZED FEEDBACK
Review the quadratic formula. Identify a, b, and c, then substitute each value carefully. Pay attention to negative signs.

STUDENT OUTCOME
Improved

RETENTION
New student profile and outcome saved.
Updated case base: 5 cases

CBR CYCLE COMPLETED
Retrieval -> Reuse -> Revision -> Retention
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REFLECTION

The project involved developing a Mathematics Intelligent Tutoring System that uses Case-Based Reasoning to assist students with repeated mathematical errors. The case base was organized to include previous problems, error patterns, interventions, and learning outcomes. One challenge was creating a similarity method that could identify which previous student case was most related to the new problem. This was addressed by comparing the mathematics topic and the repeated error pattern of each case. Another challenge was designing feedback that was not only retrieved from an old case but also adapted to the student's specific learning difficulty. Through this activity, I learned that CBR uses previous experiences to solve new problems through retrieval, reuse, revision, and retention. I also learned that saving new student profiles, personalized feedback, and outcomes allows the tutoring system to improve its recommendations for future cases.